

# XRLive: A Vertically Integrated Project (VIP) Using Extended Reality in Live Performance to Motivate Interdisciplinary Education and Scholarship

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Figure 1: Avatars, guided by human counterparts, performing live on a virtual stage

## Abstract

In this talk, we describe the XRLive Project, a cross-discipline, experiential learning opportunity that focuses on the production of live musical, theatrical, and dance performances using advanced technologies such as VR/AR/XR and motion capture. We utilize a Vertically Integrated Project (VIP) approach to recruit and engage students from different majors and disciplines. We will discuss the logistics of the project and share conclusions on how use of the VIP structure has contributed to its success.

## CCS Concepts

• **Applied computing** → **Collaborative learning; Performing arts; • Computing methodologies** → **Mixed / augmented reality; Virtual reality.**

## Keywords

vertically integrated projects, AR/VR, interdisciplinary learning

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## 1 Introduction

Over the past decade, the Computer Science Department and the School of Design at Rochester Institute of Technology have implemented a collaborative learning project called Virtual Theater. The goal of the project is to enable live performance in a virtual space with participants in different physical locales [Schweppe and Geigel 2011]. Our overall intention has always been to engage students by promoting interdisciplinary collaboration between artists and technologists, each focusing on their own expertise and talent contributing to a larger-scale and common project.

The project has produced several live theater and dance presentations that combine mixed reality, projections, and motion capture, all within the collaborative spirit of the original project [Dining and Geigel 2020; Geigel et al. 2011, 2023]. However, creating these productions has not been without its challenges. Although we were able to successfully assemble student teams for each production, ad hoc methods of recruitment that relied on informal faculty connections and word-of-mouth advertising were employed. Crediting students for their work was just as challenging, as we used a mix of assignments from related classes, independent studies, and capstone projects to provide students credit for their participation. Finally, the project lacked continuity from term to term as when graduating

students leave, they take their expertise on the project with them, forcing the project to start from scratch for future productions.

## 2 Vertically Integrated Projects

To address these issues, we recently adapted the management structure of the project to a Vertically Integrated Project (VIP) approach.

In short, the VIP approach enables students to participate in long-term, faculty-led research and creative projects and receive academic credit for doing so. The method is centered on the creation of specialized, repeatable courses in which students are enrolled and for which they gain credit for their participation in the project.

The initiative dates back to the early 2000's. Developed at Purdue University [Jan Allebach and Krueger 2006], the initiative has grown into a global consortium, currently comprising 50 participating universities from over 14 countries [Inc. 2024]. RIT joined this consortium in the summer of 2023.

The VIP model has eight essential elements [Inc. 2024]:

- (1) Projects are embedded in faculty mentor's scholarship and exploration.
- (2) Projects are long-term and large-scale, continuing for many years, even decades.
- (3) Program is curricular and all participating students are graded.
- (4) Students can participate and earn credits toward their degrees for at least two years.
- (5) Learning outcomes focus on the development of both disciplinary and professional skills.
- (6) Multi-disciplinary teams are encouraged but not required.
- (7) Dedicated classroom and meeting spaces.
- (8) Faculty/student participation is based on mutual interest

## 3 The XRLive VIP

Expanding upon the original goal of Virtual Theater, we defined the XRLive VIP; a cross-discipline, experiential learning opportunity focusing on live musical, theatrical, and dance performances using advanced technologies such as VR/AR/XR and motion capture.

Students enrolling in the VIP explore academic areas that span the technical (VR/AR systems, computer vision), the artistic (asset and character design, lighting), and the use of production tools (audio, motion capture). In addition, students are able to hone their soft skills and collaboration abilities.

All work performed as part of the VIP contributes to the presentation of a live experience shown at a public forum at the end of the term. For the 2024-5 academic year, we continue our exploration of the original virtual theater vision, placing a particular focus on the integration of body and facial motion capture to guide avatars on a virtual stage (Figure 1). In addition, some students worked on a side project that incorporated live motion capture effects with a dance show that was being performed on campus.

During the 2024-2025 academic year, 13 students enrolled in the VIP in the fall and 17 in the spring. The students came from a variety of majors and years (Table 1), with 5 of the 13 Fall students rejoining the VIP in Spring.

## 4 Discussion

Overall, we found the VIP structure to greatly enhance the management of the XRLive project; addressing the previously mentioned

**Table 1: Student enrollment by major and year - numbers in parenthesis indicate number of returning students**

| Major                   | Fall 2024 | Spring 2025 |
|-------------------------|-----------|-------------|
| Computer Science        | 7         | 9 (3)       |
| Game Design             | 1         | 3           |
| New Media Design        | 2         | 0           |
| Individualized Studies  | 2         | 1 (1)       |
| Motion Picture Sciences | 1         | 1 (1)       |
| Media Arts / Tech       | 0         | 1           |
| Computer Engineering    | 0         | 1           |
| Software Engineering    | 0         | 1           |

  

| Year         | Fall 2024 | Spring 2025 |
|--------------|-----------|-------------|
| Sophomore    | 4         | 4 (2)       |
| Junior       | 1         | 1 (1)       |
| Senior       | 4         | 6           |
| Grad Student | 4         | 6 (2)       |

challenges. The VIP initiative, now being endorsed by the Institute, provides a common recruitment vehicle as they are advertised on the Institute Web site and shared with student advising staff. Furthermore, the VIP helped in building of interdisciplinary teams as the VIP courses are open equally to all majors. Assigning credit for student work is central to the VIP philosophy, as student receive VIP course credits for their participation, a course that will show up on their transcript and contribute to their program of study just like any other elective course. Students from the fall who repeated the course in the spring provided project continuity allowing us to start immediately on subtasks in the new term. Though we have no data on overlap in multiple academic years, it is our hope that students, particularly sophomores and juniors, will choose to return to the project next Fall.

Feedback from students has been overwhelmingly positive with many of the comments highlighting teamwork and multidisciplinary collaboration. Several comments also focused on how the VIP offered something completely different from what they would have pursued on their own.

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